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Keyboard & Interfaces

Power

Processor

File: BIT90_26_Keyboard_Interfaces.kicad_sch

File: BIT90_26_Power.kicad_sch

File: BIT90_26_Processor.kicad_sch

H1
MountingHole

H2
MountingHole

H3
MountingHole

H4
MountingHole

H5
MountingHole

H6
MountingHole

Audio & Video

Controllers

File: BIT90_26_Audio_Video.kicad_sch

File: BIT90_26_Controllers.kicad_sch

>> WORK IN PROGRESS <<

>> PROTOTYPE <<

Based on original schematics from BIT Corporation (普澤有限公司)
Modern recreation of the BIT90

Brett Hallen

Sheet: /
File: BIT90_26_Prototype.kicad_sch

Title: BIT90/26 Computer

Size: A3

Date: 7/JUN/2026

Rev: PROTOTYPE

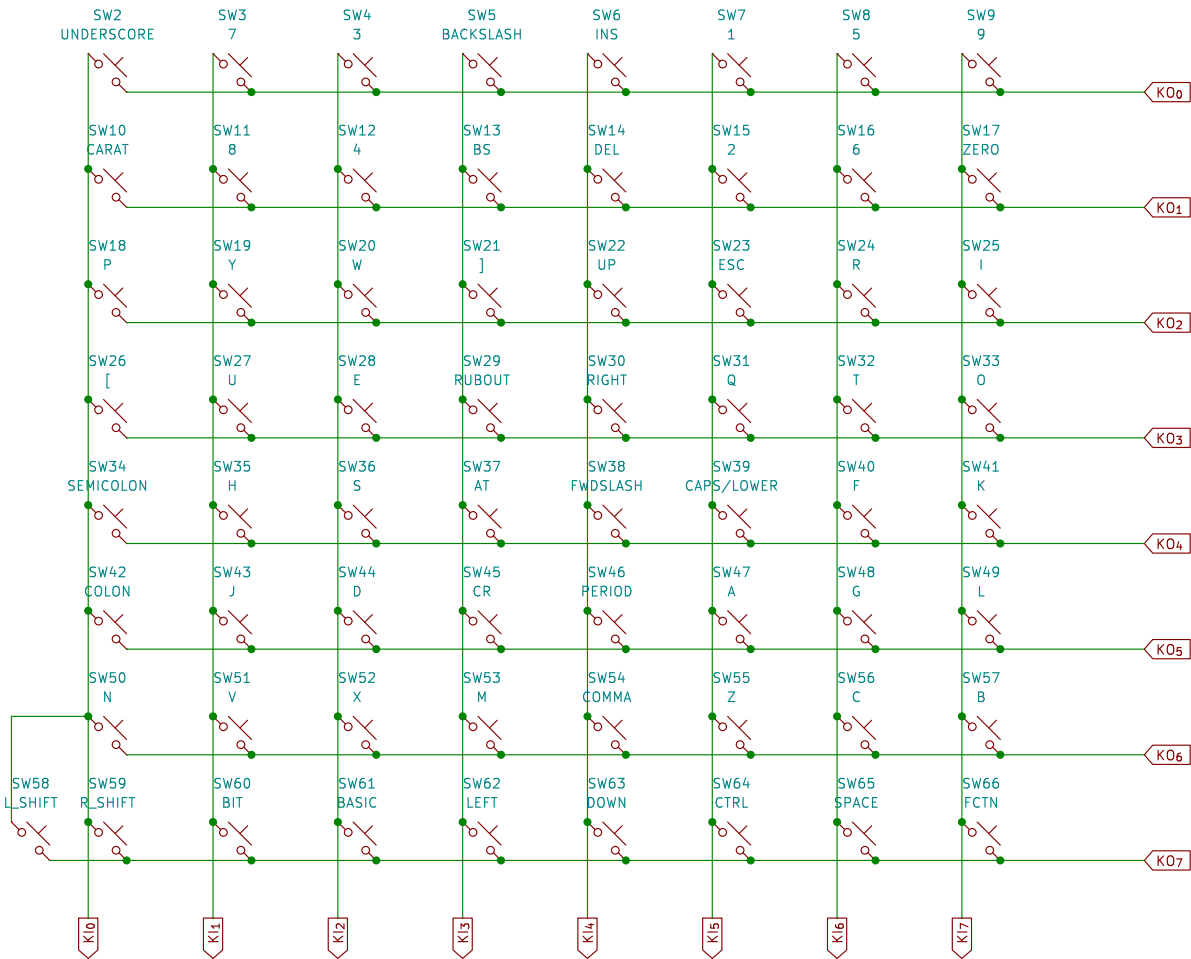
KiCad E.D.A. 9.0.9

Id: 1/6

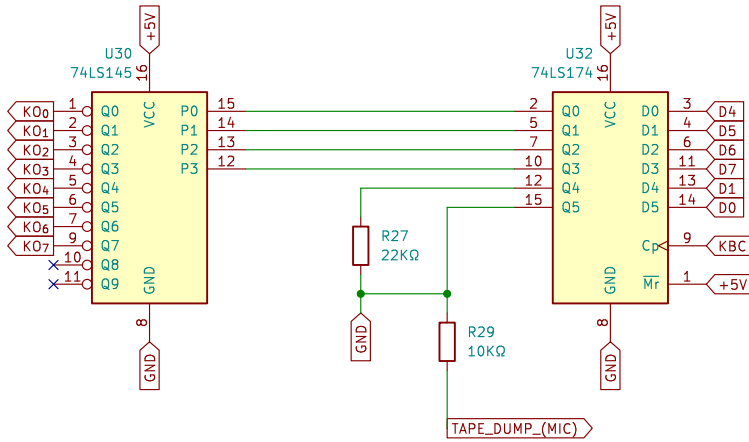
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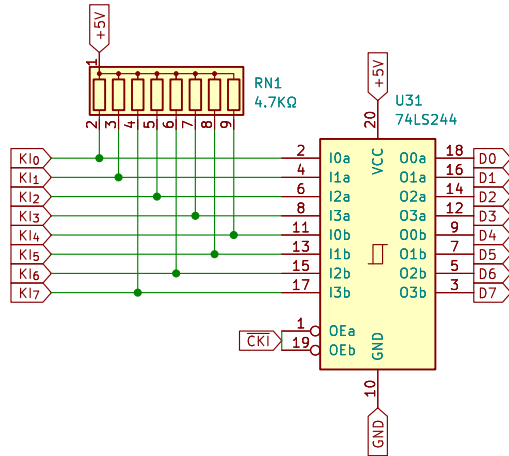
Keyboard Matrix



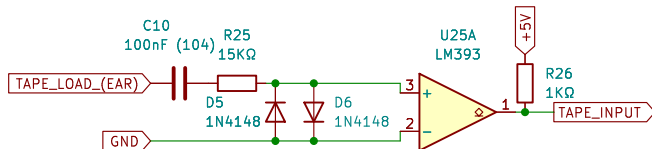
Keyboard Row Outputs
Tape Dump



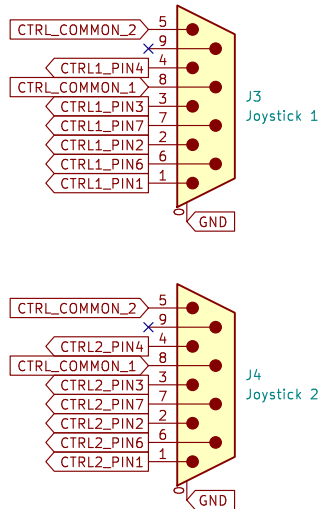
Keyboard Column Inputs



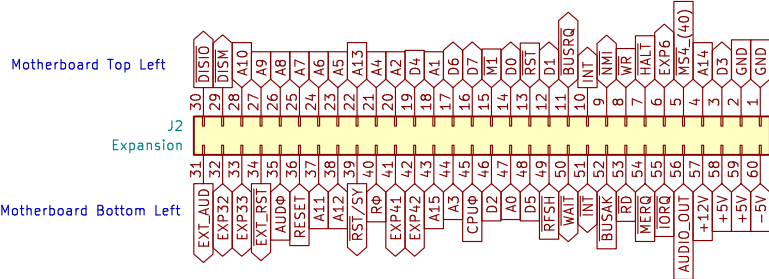
Tape Load



Joysticks
Doesn't match ColecoVision as far as I can tell

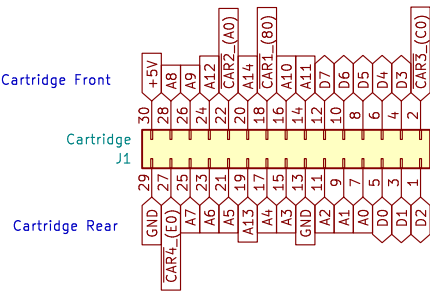


System Expansion (Edge Connector)



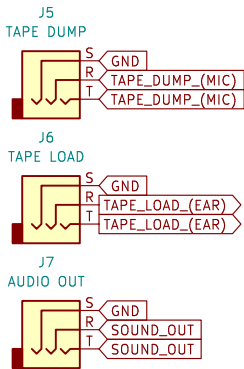
Pin 6 = EN_20 on CV
On BIT90 we have BA2 instead
thus is not connected here

ColecoVision Cartridge

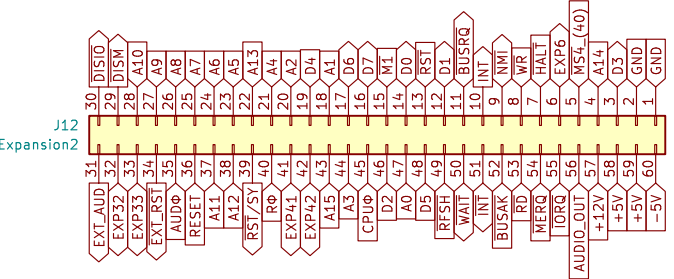


Tape & Audio
On BIT90 these are combined in a DE9 plug along with composite video

Audio Output: 1.5Vp-p
Tape Load (EAR): 500mVp-p
Tape Dump (MIC): 90mVp-p
Video Output: Dedicated HDMI board (Pico9918)



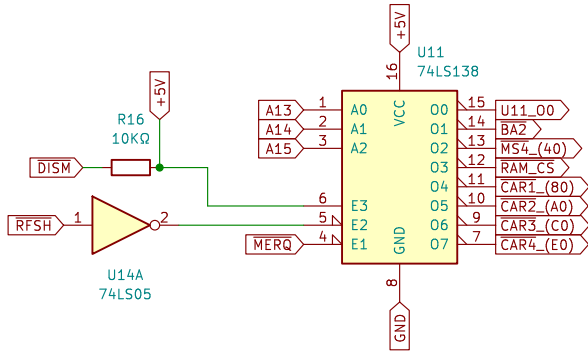
System Expansion (Pen Header)
Additional option for connecting/accessing the expansion bus



These expansion bus pins aren't connected anywhere else at the moment
EXTVE (pin 32)
EXTV (pin 33)
I will pass any unconnected pins through to the extra 2x30 pin header in case they are needed in future.
BUSRQ wasn't connected to Z80, just pulled high, so I have connected it through.
INT wasn't connected either, just INT via an inverter so I have connected it through.

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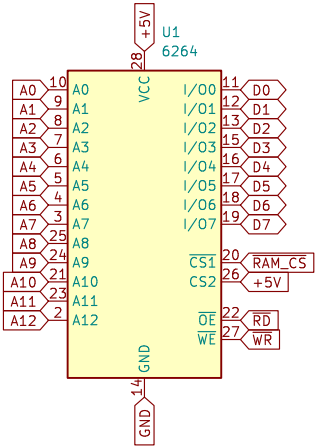
Chip Select



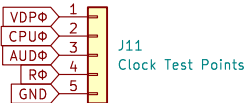
M0	ColecoVision BIOS
BA1 & BA2	BASIC
RAM_CS	System RAM
CAR1 to CAR4	Cartridge
MS4	Expansion

M0 & BA1 BEHAVIOUR
A13 = H
A14 = H
A15 = H
M0 ASSERTS WHEN A3 = L
BA1 ASSERTS WHEN A3 = H
Y1 IS STROBE

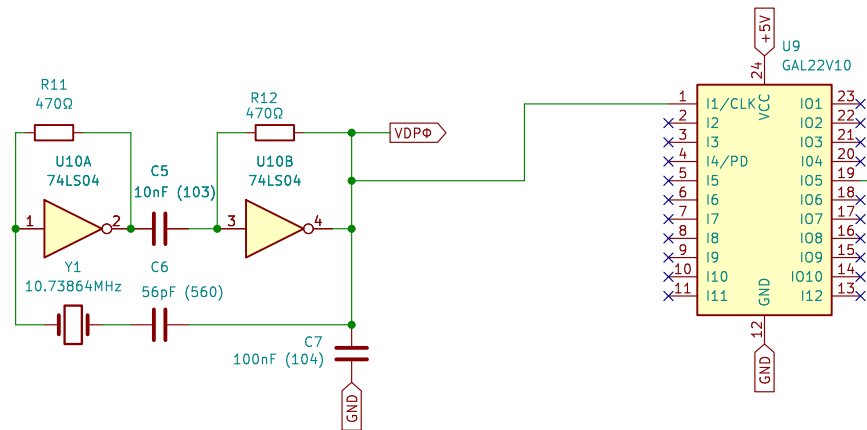
System RAM
BIT90 had 2KB originally in an 8KB block
Let's upgrade to 8KB
Upgrading further to 32KB will require additional logic
Let's see if this works first ...



Clock Test Points

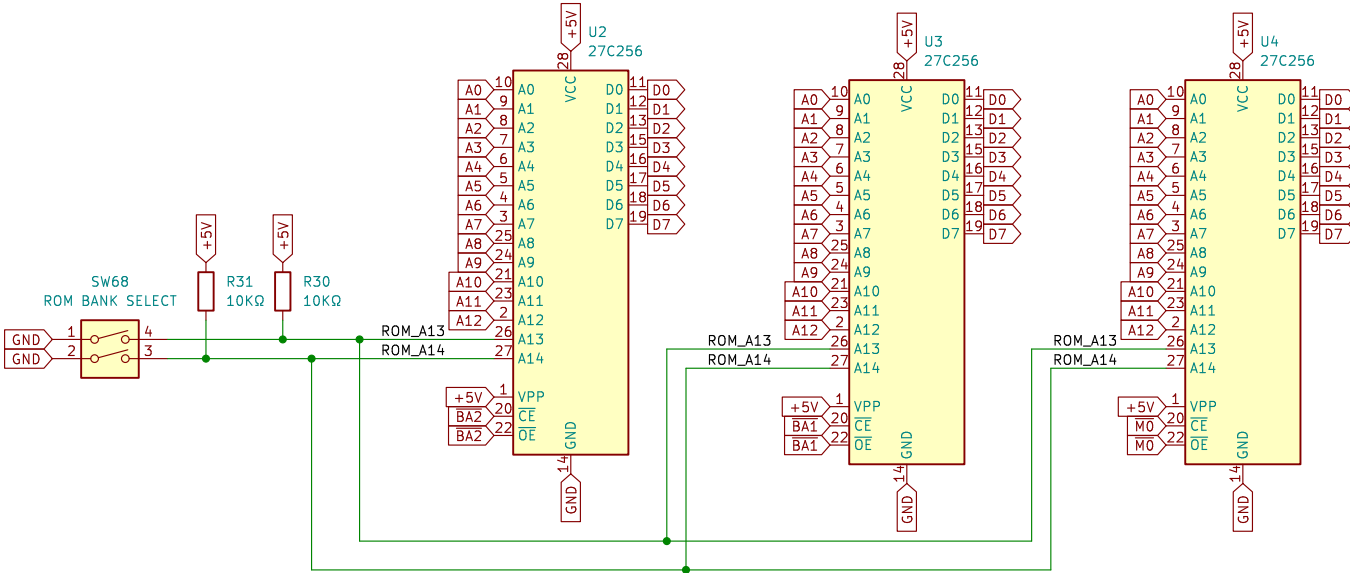


System Clocks



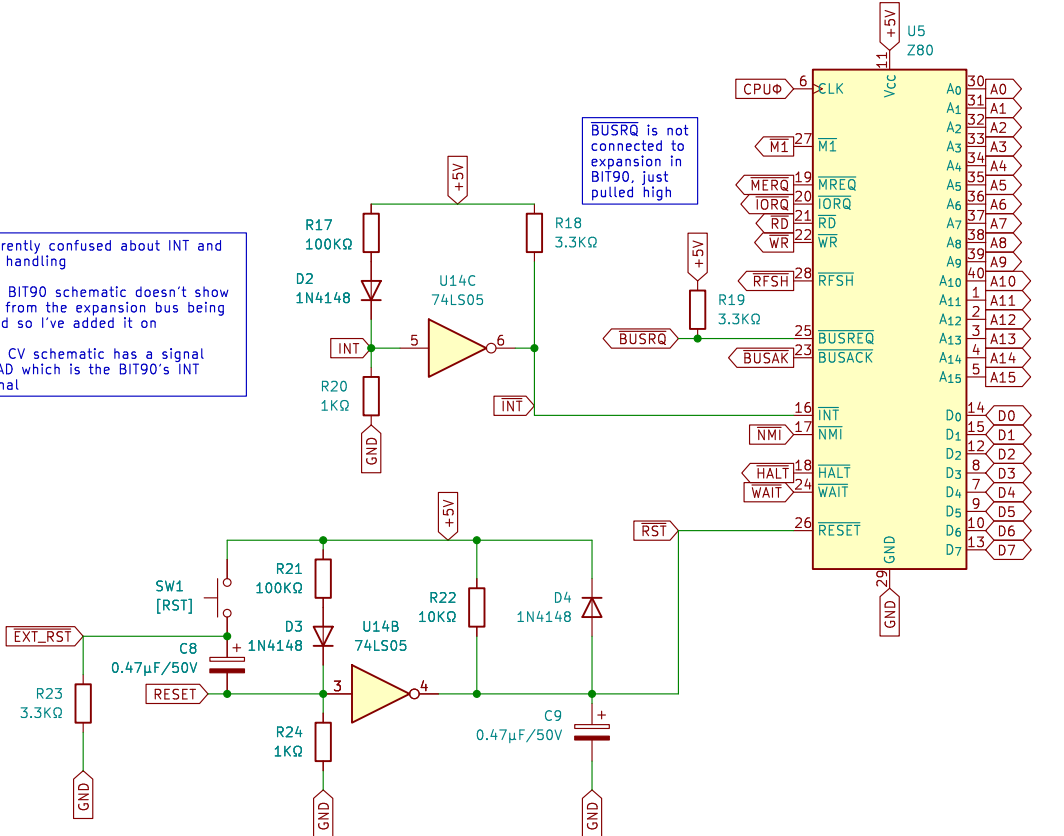
CPUΦ = CPU clock (3.57MHz)
VDPΦ = video clock (10.73864MHz)
AUDΦ = audio clock (called EXTΦ on BIT90)
RΦ = current-limited CPU clock to expansion

System ROMs
BASIC1, BASIC2, ColecoVision BIOS
Original ROMs are 8KB
Using 27C256 allows four banks that can be switched



CPU & Reset Circuit

Currently confused about INT and INT handling
The BIT90 schematic doesn't show INT from the expansion bus being used so I've added it on
The CV schematic has a signal QUAD which is the BIT90's INT signal



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